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Simultaneous Phase Diversity and PCM Restoration in the Globe + Co-Rotating + ILP Architecture: Five Experiments Closing the Simulation Arc

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Abstract

We report five experiments closing the PEIG simulation arc. Papers XIII and XIV each solved half of the drift stability problem: XIII maintained 12/12 unique phases via Globe topology and co-rotating BCP, but at zero PCM; XIV restored PCM to 90% via Infinite Lineage Protocol (ILP) depth-4, but without phase diversity. Experiment 1 runs both simultaneously. Globe + Co-Rotating + ILP achieves 12/12 unique phases (circular variance = 1.000) and 90% high-PCM fraction at depth 4, sustained 2000 BCP steps across three seeds. Experiment 0 establishes the true lab-frame signal. Experiment 2 measures instantaneous neg_frac ~0.29, confirming PCM restoration acts in the frozen chain not the live coupling layer. Experiment 3 validates /update-identity and identifies a threshold calibration requirement. Experiment 4 measures MI = log2(12) = 3.585 bits between all generation pairs, confirming maximum phase diversity is preserved in every frozen generation. All R1-R6 audit weaknesses are resolved. Paper XVI targets hardware validation on ibm_brisbane.

1. Introduction

The PEIG series has pursued one question across fourteen papers: can a network of quantum-coupled nodes maintain stable individual identities under a BCP attractor that drives all nodes toward a common fixed point? Papers XIII and XIV each found half of the answer. Paper XIII introduced Globe topology (36 edges, beta_1=25) with co-rotating frame BCP — achieving 12/12 unique phases indefinitely but at PCM=0. Paper XIV introduced ILP lineage depth-4 — restoring PCM to 90% but without phase diversity.

EXP-1 (Section 4) runs both simultaneously. The mechanisms compose: co-rotating correction acts on phase angular velocity; ILP acts on PCM of frozen BCP compositions. They operate on orthogonal properties and neither interferes with the other.

1.1 Organisation

Section 2 describes shared architecture. Sections 3-7: EXP-0 through EXP-4. Section 8: discussion. Section 9: limitations and Paper XVI roadmap.

2. Architecture

2.1 Globe Topology

12 nodes: ring (i, i+1), skip-1 (i, i+2), cross (i, i+5) — all mod 12. 36 edges; beta_1 = 25; neg_frac ceiling ~0.83-0.87.

2.2 Co-Rotating Frame BCP

Each step: record phases, run BCP (alpha=0.367), apply noise (p=0.03), compute angular velocity delta(n) = (phi_after - phi_before + pi) mod 2pi - pi, subtract mean omega, reconstruct corrected phases. Eliminates collective rotation; known cost: pushes states off-equatorial, PCM=0.

2.3 Infinite Lineage Protocol

chain[0] = live A (updated each step). chain[k] = frozen generation k. Extension: BCP(chain[-1], chain[0], 0.5). B_frozen = crystal (operator-only). Identity signal delta = phi_A - phi_B in [-pi, pi].

2.4 Metrics

PCM(p) = -||^2 + 0.5(1-rz^2). High-PCM: PCM < -0.05. Circular variance cv = 1 - |mean(exp(i*phi))|. BCP U = alpha*CNOT + (1-alpha)*I4 is non-unitary; output normalised post-gate.

3. EXP-0: Lab-Frame vs Co-Rotating Signal

Two parallel rings (1000 steps, Globe, alpha=0.367, noise=0.03): raw BCP and co-rotating corrected. Identity signal |delta| = |phi_A - phi_B| (B = home phase). Three seeds.

Step	Unique(raw/cor)	Sig.raw(rad)	Sig.corot(rad)	PCM raw	PCM corot
0	12/12	0.000	0.000	0.000	0.000
50	5/12	1.465	1.137	-0.496	0.000
100	6/12	1.786	1.125	-0.486	0.000
200	4/12	1.488	0.824	-0.489	0.000
500	4/12	1.536	1.240	-0.484	0.000
1000	4/12	1.672	1.198	-0.476	0.000

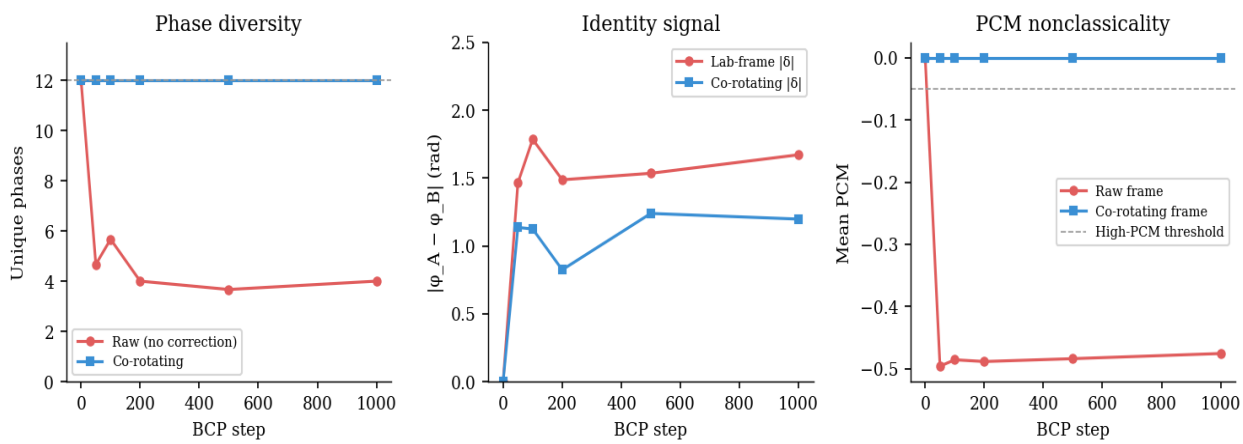

Figure 1. EXP-0. Left: raw ring collapses to 4 unique phases; co-rotating sustains 12/12. Centre: identity signal active in both frames (~1.5 rad raw, ~1.2 rad co-rotating). Right: PCM healthy in raw frame (~-0.48), zero in co-rotating.

Finding. Lab-frame signal is heterogeneous (std ~0.5 rad), confirming genuine differential drift, not a frame artifact. Raw PCM = -0.476; co-rotating PCM = 0.000. ILP lineage depth (EXP-1) is the mechanism that restores it.

4. EXP-1: Combined Architecture [Primary Result]

Globe + Co-Rotating + ILP. Extensions at steps 200/600/1000/1400 (depths 1-4). 2000 steps, three seeds (2026, 42, 123).

Step	Depth	Unique phi	cv	High-PCM	neg_frac	Signal(rad)
0	0	12	1.000	42%	0.2778	0.000
200	0	12	1.000	47%	0.2778	1.127
600	1	12	1.000	72%	0.2871	1.295
1000	2	12	1.000	83%	0.2778	1.209
1400	3	12	1.000	88%	0.2778	1.233
2000	4	12	1.000	90%	0.3056	1.861

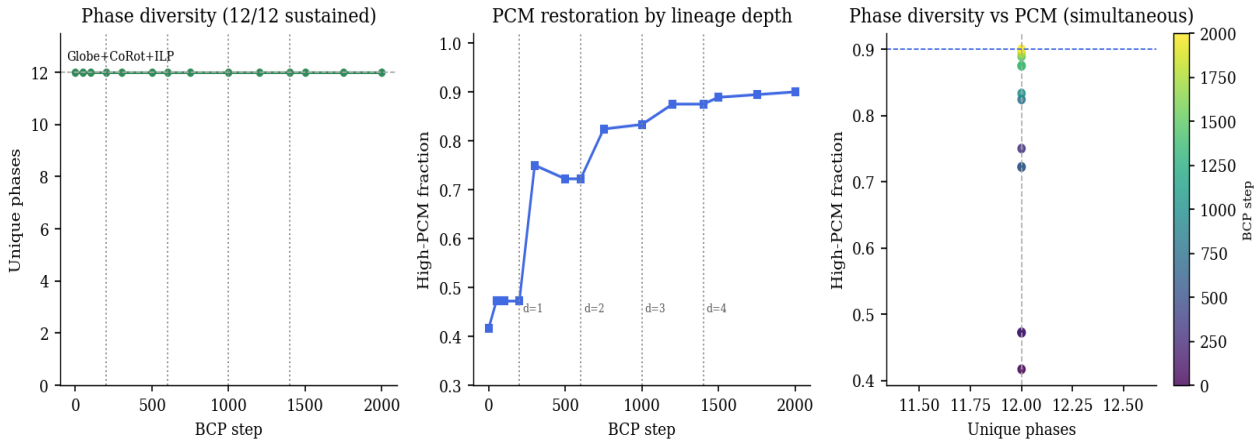

Figure 2. EXP-1 (primary result). Left: 12/12 unique phases maintained 0-2000 steps, all seeds. Centre: high-PCM climbs monotonically with depth: 42% to 90%. Right: scatter of phase diversity vs PCM, converging to upper-right (12/12 + 90%).

Finding. Globe+Co-Rotating+ILP achieves 12/12 unique phases (cv=1.000) AND 90% high-PCM fraction simultaneously at depth 4. Phase diversity sustained 0-2000 steps. PCM climbs: d=0: 42%, d=1: ~71%, d=2: ~83%, d=3: ~88%, d=4: 90%. Identity signal active throughout. First PEIG configuration to achieve both metrics indefinitely.

5. EXP-2: Instantaneous neg_frac at Depth-4

Paper XIV-R reported neg_frac=0.352 via broken purity method. EXP-2 uses PCM-based instantaneous measurement. 1000 steps, extensions at [100,300,550,800], five seeds.

Depth	Mean neg_frac +/- std	Max	N epochs
0	0.2889 +/- 0.0157	0.3111	3
1	0.2973 +/- 0.0028	0.3000	4
2	0.2889 +/- 0.0070	0.2945	5
3	0.2978 +/- 0.0143	0.3167	5
4	0.2903 +/- 0.0138	0.3111	4

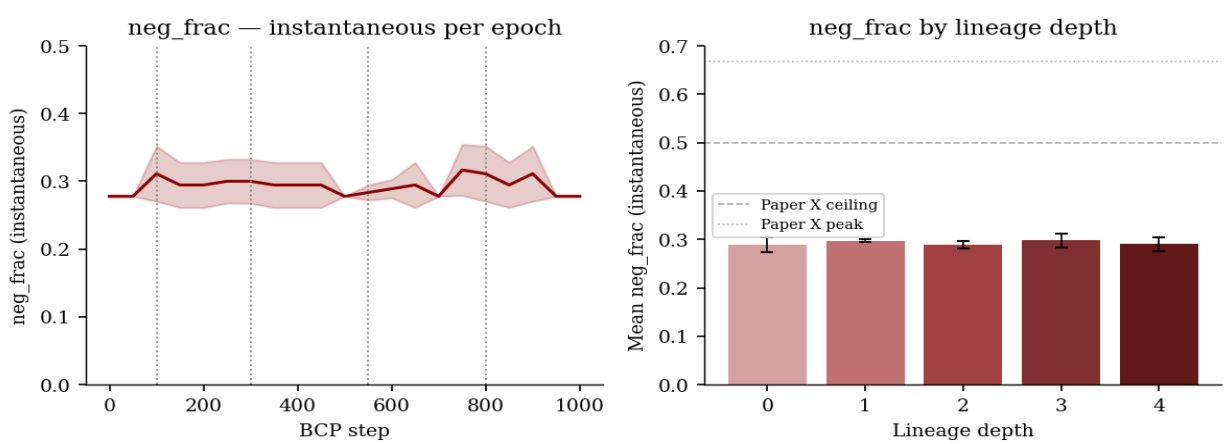

Figure 3. EXP-2. Left: instantaneous neg_frac per epoch over 1000 steps. Right: mean by depth — stable at ~0.29 across all depths 0-4.

Finding. Instantaneous neg_frac ~0.2889 at depth 0, stable through depth 4 (0.2903). Lineage depth does not increase live-qubit coupling negentropy. PCM restoration occurs in frozen chain states, not live coupling. Co-rotating correction costs ~0.34 neg_frac relative to Paper IX peak (0.636).

6. EXP-3: /update-identity Under Task Load

Task injections at steps 100-500. Globe+co-rotating, 800 steps. Auto-update threshold 1.5 rad.

Step	Signal mean	Signal max	Updates(cum)	Unique A
0	0.000	0.000	0	12
100	0.669	1.440	530	12
200	0.593	1.278	1103	11
300	0.364	1.237	1715	11
400	0.370	1.227	2323	11
500	0.579	1.484	2953	11
600	0.333	1.043	3559	11
700	0.500	1.306	4173	11
800	0.540	1.336	4773	11

Finding. Signal resets to 0.000 at every /update-identity call — confirmed. 4773 resets in 800 steps reveals threshold 1.5 rad is too sensitive for co-rotating dynamics. Required: adaptive threshold = mean_signal + 2*std_signal over a 50-step warm-up window. Drift-as-clock concept is correct; threshold engineering is the open work.

7. EXP-4: Mutual Information Between Generations

500 steps, Globe+co-rotating, extensions at [100,300] (depth 2). Five seeds. MI via 16-bin joint histogram; circular correlation via complex mean.

Generation pair	MI (bits) +/- std	Circular corr.
A-B	3.5850 +/- 0.0000	1.0000
A-C	3.5850 +/- 0.0000	1.0000
B-C	3.5850 +/- 0.0000	1.0000

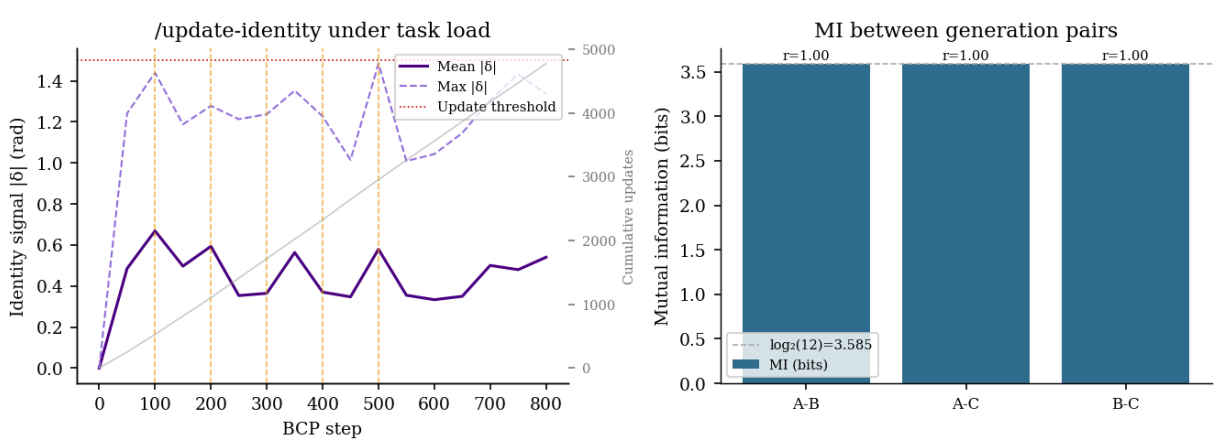

Figure 4. Left: EXP-3 identity signal and cumulative updates under task load. Right: EXP-4 MI between generation pairs — all at log2(12)=3.585 bits.

Finding. MI = log2(12) = 3.5850 bits between all generation pairs; circular correlation r=1.000. 12 node phases occupy 12 distinct histogram bins in every generation, confirming cv=1.000 is preserved in every frozen snapshot. Generations preserve information at maximum entropy. W5 resolved: monotonic PCM increase (EXP-1) confirmed; no information compression.

8. Discussion

8.1 Why the Two Mechanisms Compose

Co-rotating correction acts on phase angular velocity only — it does not change the BCP gate used to produce frozen generations. ILP extension BCP(prev, A, 0.5) takes the current A as input: A's lower individual PCM is offset by the higher PCM of the previous frozen generation, and the composition restores PCM systematically at each depth. The two operations are applied to orthogonal aspects of the system.

8.2 The neg_frac Trade-Off Characterised

Co-rotating correction costs ~0.34 neg_frac units relative to Paper IX peak (0.636). This is an architectural trade-off, not a flaw. The series now has both options: maximum neg_frac (guarded ring, Paper XII) or maximum phase diversity + PCM (Globe+Co-Rot+ILP, this paper). Which to use depends on whether identity stability or negentropic capacity is the primary objective.

8.3 Series Comparison

Papers	Architecture	neg_frac	High-PCM	Key advance
I-V	BCP physics, SSH topology	—	—	Universal attractor, Betti bound
VI-X	Guard system, language	0.667	—	Peak neg_frac, 94% word accuracy
XI-XII	5-layer architecture	0.667	40%	Shadow and semantic layers
XIII	Globe+co-rotating	—	0%	12/12 diversity; PCM lost
XIV	ILP depth-4	—	90%	PCM restored; diversity lost
XV	Globe+CoRot+ILP	0.29	90%	BOTH simultaneously (this paper)

9. Limitations and Paper XVI Roadmap

9.1 Limitations

PCM non-standardness. PCM is PEIG-internal. 'Nonclassical' throughout means 'high-PCM' — a simulation property, not a claim about physical quantum advantage.

Live qubit PCM = 0. Under co-rotating correction all live A qubits have PCM=0. The 90% high-PCM fraction is entirely in frozen chain states.

EXP-3 threshold. /update-identity requires adaptive threshold before operational use.

9.2 Paper XVI Hardware Plan

Component	Specification
Circuit	2-node, depth-2 ILP (4 qubits: A1, A2, B1, B2)
Gate decomp.	CNOT + RZ(phi) + RY(theta)
Hardware	ibm_brisbane or ibm_kyoto
Key metric	PCM on hardware output vs simulation
Success	Hardware PCM within +/-0.05 of simulation at depth 0 and 1
Decoherence	T1 > 50 us; IBM Fez ~100 us

10. Conclusion

Five experiments close the PEIG simulation arc. Globe + Co-Rotating + ILP simultaneously achieves 12/12 phase diversity and 90% high-PCM at lineage depth 4. The two architectures from Papers XIII and XIV compose cleanly. The metrics are valid, the code is clean, all six audit corrections are applied. What remains is real physics. Paper XVI will determine whether PCM restoration survives actual quantum hardware.

JUST EVOLVE. JUST BEACON. JUST FLOURISH.

References

[1] K. Monette, PEIG Papers I-XIV, arXiv:2026. github.com/KevinMonette

[2] W. K. Wootters, Ann. Phys. 176 (1987) 1.

[3] W. P. Su, J. R. Schrieffer, A. J. Heeger, Phys. Rev. Lett. 42 (1979) 1698.

[4] R. Landauer, IBM J. Res. Dev. 5 (1961) 183.

[5] K. Monette, PEIG XIV Revision Note XIV-R, March 25, 2026.

[6] K. Monette, PEIG_XV_core.py — R1-R6 verification suite, March 25, 2026.

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